

Invertebrate community dynamics and insect emergence in response to pool drying in a temporary river

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SUMMARY

1. Previous studies of invertebrate responses to hydrological disconnection from the flow and drying have focussed on habitats with long hydroperiods (dry slowly after disconnection, i.e. months to years). We focus here on the effects of short hydroperiods (dry rapidly after disconnection, i.e. hours to days) on river invertebrate communities and insect emergence.
2. We used natural scour pools in an alluvial river to measure changes in the invertebrate benthos and insect emergence during a 5-month drought. The scour pools represented a gradient of hydroperiods (time from disconnection to complete water loss). We monitored water level and physicochemical conditions continuously. Benthic invertebrates on the bottom of each pool and aquatic insects that emerged from the surface of each pool as aerial adults were sampled at 1- to 7-day intervals during connected, disconnected and dry periods. We used the data on water level to group the pools into quick-drying and slow-drying hydroperiod classes.
3. Community structure changed as pools disconnected and dried. The densities of larval Diptera were higher in the connected period than in the disconnected period, while those of larval Ephemeroptera and Trichoptera pupae were higher in the disconnected period. Taxon richness was higher in quick-drying than in slow-drying pools. Shannon–Wiener diversity increased as water level decreased in slow-drying pools, but there was no detectable effect of water level on diversity in quick-drying pools.
4. Total emergence of aquatic insects increased with decreasing water level in the scour pools. However, emergence rates (individuals per day) of Ephemeroptera and the numerically dominant genus *Deleatidium* decreased with declining water level.
5. Some aquatic insects did not complete development in quick-drying pools. Benthic invertebrates that persisted in pools until complete drying were dominated by taxa with desiccation-resistance traits.

Keywords: benthic invertebrates, drying, emergence, hydroperiod, intermittent river, scour pool

Introduction

Temporary rivers are characterised by the repeated onset and cessation of flow, alternating emersion and immersion of benthic habitat, and hydrological connection and disconnection of pools, backwaters and floodplain ponds (Larned *et al.*, 2010a). In river reaches with a number of pools, loss of flow initiates a sequence of state changes in the benthic habitat (Fig. 1): (i) chains of interconnected pools appear as submerged areas as the channel contracts; (ii) connections between pools are reduced to small rivulets; (iii) rivulets dry and pools disconnect;

(iv) water levels in disconnected pools decline due to seepage and/or evaporation; and (v) disconnected pools dry completely.

Between the stages of channel contraction and pool disconnection, mobile aquatic organisms may follow the receding water into pools (Stanley *et al.*, 1994; Labbe & Fauch, 2000; Magoulick & Kobza, 2003). When pool disconnection occurs, the trapped aquatic organisms comprise discrete biotic assemblages that may differ stochastically in initial taxonomic composition and population densities (Capone & Kushlan, 1991; Lake, 2003).

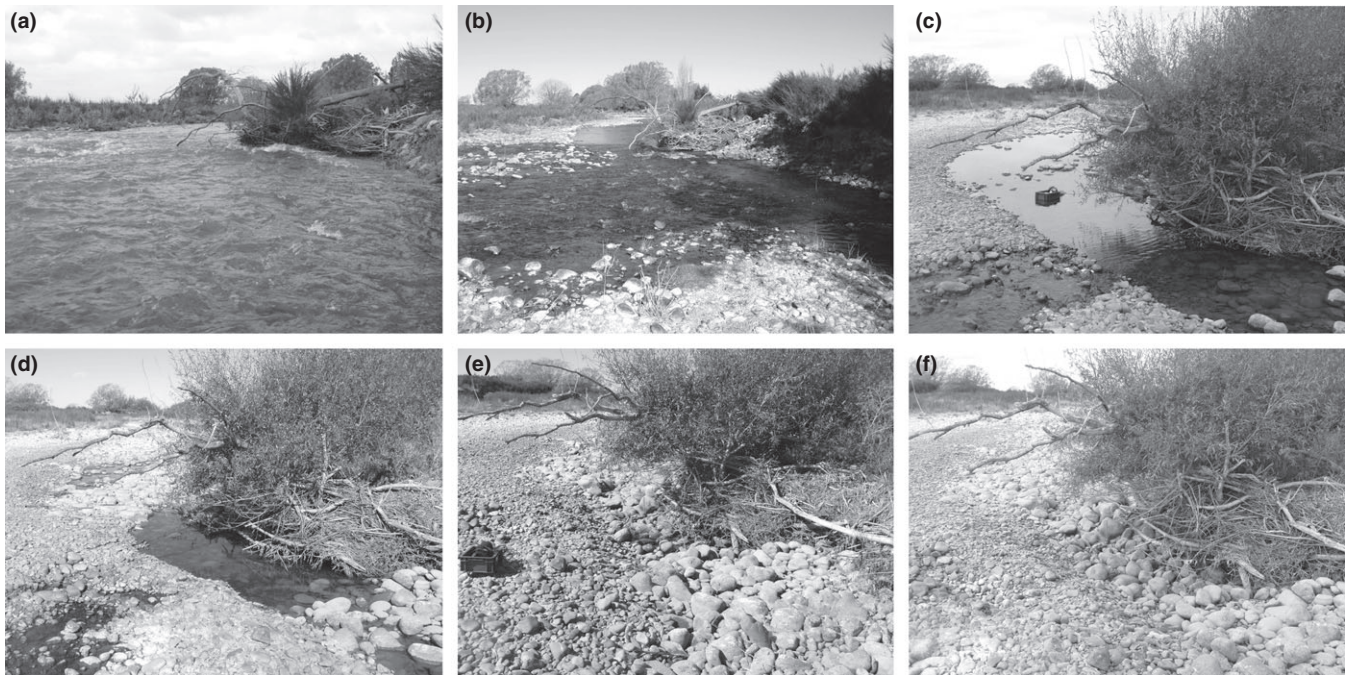


Fig. 1 Time series of habitat changes during a drying event in the Selwyn River, in the reach used for the current study. Photographs were taken at Pool 4 (see Fig. 2) between 4 December 2006 and 19 February 2007. (a) Bankfull, active channel submerged. (b) Chains of connected pools, wetted channel contracting. (c) Near-disconnected. (d) Disconnected, pools contracting. (e) Near-dry, with 1–2 cm of water under cobles. (f) Dry.

As disconnected pools contract, organism density increases, which may increase predation pressure and competition for space and food (Boulton, 1989; Lake, 2003; Acuña *et al.*, 2005; Bonada *et al.*, 2006). Pool contraction is generally accompanied by reductions in dissolved oxygen (DO) concentration, increases in diurnal water temperature, and changes in the amplitudes of diel fluctuations in DO, pH and temperature (Acuña *et al.*, 2005; Williams, 2006). These physicochemical changes are lethal to some invertebrates, but they also serve as proximal cues that elicit escape and resistance responses (Robinson & Buser, 2007; Lytle, 2008; Lytle, Olden & McMullen, 2008).

Escape and resistance responses include physiological (e.g. aestivation, anhydrobiosis) and morphological changes (e.g. pupation, metamorphosis and emergence of aerial adult insects) (Humphries & Baldwin, 2003). The minimum times required for these responses are set by maximum rates of tissue differentiation, ecdysis, synthesis of protective biomolecules and other developmental and physiological processes (e.g. Watanabe, 2006). For some aquatic invertebrates, responses to pool drying may be constrained by their mode of development. For example, holometabolous aquatic insects can only escape from drying pools as adults when the time from disconnection to complete water loss in a pool (its

'hydroperiod') exceeds the minimum time required for larval growth, pupation and emergence (Jannot, Wissinger & Lucas, 2008). Hemimetabolous insects, without a pupal stage, may be better suited to escape rapidly drying pools by emergence. Within these general modes, variation in developmental flexibility (i.e. the degree to which emergence can be decoupled from larval growth) affects the responses of individual taxa. For some aquatic insects, maturation of young, small larvae in response to drying or increased water temperature appears to increase survival in intermittent rivers (Hogg & Williams, 1996; Robinson & Buser, 2007; Brown *et al.*, 2012). In other aquatic insects, obligate larval or pupal diapause and minimum size thresholds for maturation reduce developmental flexibility (Gray, 1981; Jannot *et al.*, 2008). In pools with extremely short hydroperiods (hours to several days), accelerated development and physiological transformation cannot substantially increase emergence rates or desiccation resistance. In these cases, the taxa that survive are dominated by those with behavioural responses, such as rheotaxis and burrowing (Lytle *et al.*, 2008; DiStefano *et al.*, 2009), or constitutive traits that confer resistance to drying, including chitinous exoskeletons and respiratory spiracles (Zamora-Muñoz & Svensson, 1996; Bonada, Rieradevall & Prat, 2007).

Based on these observations, it appears that the effectiveness of escape and resistance responses in intermittent river pools is influenced by both pool hydroperiod and intrinsic properties of the invertebrates, including developmental modes and resistance traits. Time constraints, particularly in pools with very short hydroperiods, may determine which taxa emerge, resist *in situ* or die. In turn, emergence and mortality shape the composition and diversity of invertebrate assemblages remaining in disconnected pools as they dry. The taxa that persist longest in drying pools are likely to share life-history, morphological, and physiological traits that increase resistance to desiccation (Bonada *et al.*, 2007; Arscott *et al.*, 2010; Datry *et al.*, 2014).

We examined responses of aquatic macroinvertebrate assemblages to variation in hydroperiods, water level and pool state (connected versus disconnected) in the scour pools of an alluvial river reach in New Zealand. Scour pools are naturally occurring examples of habitat disconnection and contraction (Magoulick & Kobza, 2003; Corti *et al.*, 2011). Flood-prone alluvial rivers often have many scour pools with variable connectivity and patterns of water loss. Upon disconnection, these pools contain discrete aquatic assemblages that can be used as replicates in observational studies. Here we predicted that (i) pool disconnection and reduction in water level induce emergence in taxa, indicated by increased rates of emergence following disconnection and during pool drying; (ii) emergence of adult insects from pools with

short hydroperiods is dominated by hemimetabolous taxa (lacking a pupa) without a larval diapause; (iii) the taxon richness, diversity and composition of the benthos in disconnected pools change over the hydroperiod, due to differential emergence and mortality; and (iv) the benthic invertebrates remaining in drying pools are dominated by taxa with desiccation-resistance traits and holometabolous taxa with low developmental flexibility.

Methods

Study site

The Selwyn River, south-east New Zealand, rises in the foothills of the Southern Alps and flows east across the alluvial Canterbury Plains to Lake Ellesmere, on the east coast of the South Island (Fig. 2). The study area was located in the upper alluvial plains, 3–5 km downstream of the foothills–plains boundary. In the upper plains, the Selwyn River flows over poorly confined alluvial aquifers that are recharged by river seepage through a 20- to 80-m-thick vadose zone (Larned *et al.*, 2008). High seepage losses reduce river flow with distance across the upper plains, and all flow is lost for part of most years in an ephemeral section (length 45 km) that starts *c.* 3 km downstream from the foothills–plains boundary. The study reach consisted of the uppermost 2 km of this ephemeral section. It loses flow at a long-term mean rate

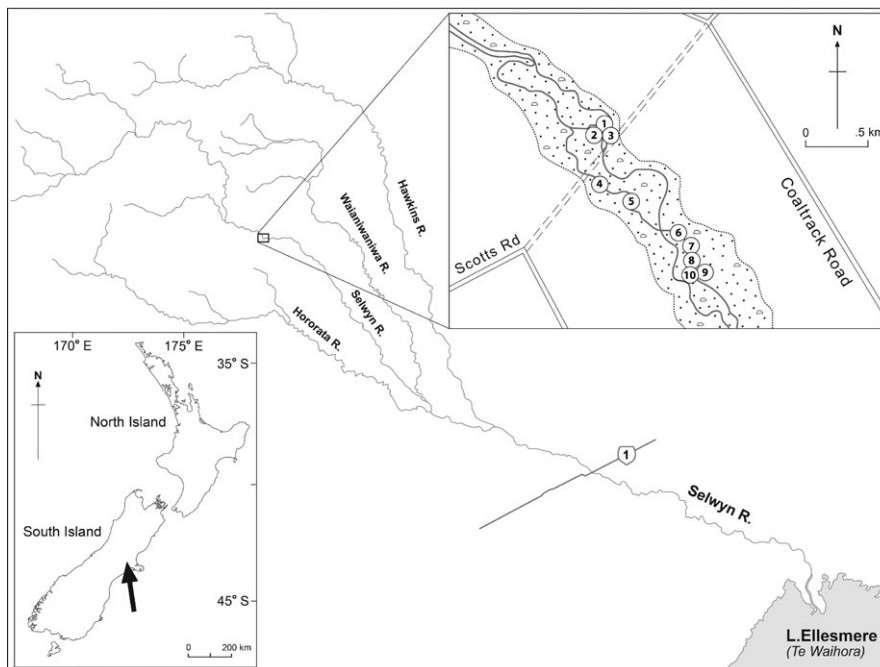


Fig. 2 The Selwyn River, New Zealand, and the study reach in the ephemeral section (right inset). Circled numbers indicate scour pools used in the study, stippled area indicates the active channel, and black lines indicate braided river channels at base flow.

of $0.3 \text{ m}^3 \text{ s}^{-1} \text{ km}^{-1}$, and the frequency and duration of zero flow increases with distance down the reach (Larned *et al.*, 2010b).

Scour pools are common in the upper Selwyn River and are created during floods by flow convergence at the confluences of channel braids, and by downward deflection of flow at large obstructions (e.g. log jams, gravel bars). The study reach generally has more than 20 scour pools, the number varying as new pools are created and existing pools are filled with sediment during bed-moving floods. The pools dry and rewet repeatedly between floods in response to flow fluctuations (Corti *et al.*, 2011). The current study took place during an extended flow recession, and all of the scour pools in the ephemeral section dried completely.

Field procedures

Ten scour pools were selected at random for the study in September 2008. The pools were spaced 5–375 m apart and ranged in maximum depth (rim to deepest point) from 0.30 to 0.95 m and in maximum water-surface area from 3 to 21 m². A capacitance water-level recorder (model ODYWL15, Odyssey Dataflow Systems, Christchurch, New Zealand) inside a screened PVC pipe was attached to a vertical steel post at the deepest point of each pool. An iButton temperature logger (Embedded Data Systems, Lawrenceburg, Kentucky) was attached to the base of each water-level recorder. Water level and temperature were recorded at 15- and 10-min intervals, respectively, and the recorders were downloaded weekly. Water temperature, DO concentration, turbidity and electrical conductivity (EC) were measured and logged at 15-min intervals in eight of the 10 study pools with Troll 9500 sensors and loggers (Troll 9500, In-Situ, Inc., Fort Collins, Colorado).

Rock baskets were used to assess changes in macroinvertebrate assemblages during pool disconnection and drying. Samples from rock baskets may not include all macroinvertebrate taxa that occur in destructive samples (e.g. Surber samples), due to limited colonisation times (Meier, Penrose & Polak, 1979). However, this limitation is offset by several advantages: rock baskets eliminate the risk of resampling previously sampled locations, reduce variability associated with sediment heterogeneity and varied sampler penetration into the sediment and reduce the need for high water flow to wash invertebrates into sampler nets. Invertebrate assemblages in the rock baskets were similar in composition to Surber samples collected in the same reach (Arscott *et al.*, 2010).

Rock baskets were made by folding 15 × 15 cm pieces of 10-mm galvanised wire fence material in half and closing the sides with plastic-coated wire. Each basket contained 16–20 basalt cobbles (33–50 mm diameter), with a total volume of c. 300 mL (measured by displacement in water). Fifteen baskets were attached end-to-end with cable ties, and the first basket in each chain of 15 was cable-tied to the steel post in each pool. The baskets were installed in the scour pools in September 2008 and left to colonise with biofilm and invertebrates for 36 days before the first sampling date. Three rock baskets were removed from the end of the basket chain in each pool on each sampling date and were not replaced.

Sampling dates were determined for each pool by the rate of drying. Samples were collected weekly during connected periods and daily during disconnected periods and after complete drying. In each pool, ≥ 3 samples (≥ 9 rock baskets) were collected during the connected and disconnected periods and ≥ 1 sample (≥ 3 rock baskets) was collected during the dry period. One pool (Pool 6) was sampled twice in 1 day (at 0800 and 1400 hours) due to rapid drying. To prevent loss of invertebrates from the baskets, a 300- μm -mesh hand net was placed around the baskets before they were detached from the chains and lifted from the pools. Baskets were processed in the field by emptying the contents into a 63- μm -mesh sieve, rinsing in filtered river water and preserving the retained material in 90% isopropyl alcohol.

Floating emergence traps were used to measure rates of aquatic insect emergence (defined as aquatic and semi-aquatic taxa leaving pools through flight) in response to pool disconnection and drying. The traps had 50 cm high triangular sides, 47 cm diameter circular bases, and wooden lids fitted with 250-mL plastic collection containers. The trap sides and bases were constructed from 22-mm-diameter PVC pipe and 300- μm -mesh nylon screen, with foam tubes on the bases for floatation. Approximately 50 mL of ethylene glycol was added to the collection containers in the trap lids as a preservative. Three emergence traps were installed in the centre of each pool on 3 November 2008, prior to pool disconnection. The sample collectors were emptied and preservative replaced weekly when pools were connected and then daily from the date of pool disconnection until drying. Insects removed from the collection containers, and the trap enclosures were preserved in 90% isopropyl alcohol. Eight of the 10 pools had continuous emergence-trap data that were used for analysis; the traps in the remaining two pools were periodically

overturned by nocturnal animals, and the data from these pools were omitted.

Laboratory procedures

Benthic invertebrates from rock baskets were counted and identified to the lowest practical taxonomic level using keys in Winterbourn, Gregson & Dolphin (2006); most insects were identified to genus or species, and chironomids, Crustacea, Acarina and terrestrial invertebrates were identified to family or order. Insect pupae in the rock baskets were also counted and identified to order. Adult insects in emergence traps were identified to species, except Diptera which were identified to family.

Each invertebrate taxon in the rock-basket and emergence-trap samples was categorised on the basis of eight life-history and morphological traits related to emergence or desiccation resistance: developmental mode, voltinism, flexibility of developmental rate, duration of larval phase, duration of pupal phase, presence of aerial adult phase, type of respiratory organ and types of morphological desiccation-resistance traits. Information about traits came from a database for New Zealand aquatic invertebrates (Freshwater Biodata Information System – FBIS, 2014) and from primary literature and personal communications with experienced invertebrate biologists. Detailed information about traits is limited for many New Zealand invertebrates, and some taxa could not be identified to the species level. In these cases, trait categories were assigned based on traits known for higher taxonomic levels. For example, if the voltinism category was unknown for a genus, but the other genera in the family were univoltine, the genus in question was assumed to be univoltine.

Data analyses

The dates and times of disconnection and complete drying in each scour pool were determined by the examination of data on water level. Pool hydroperiod was calculated as the time elapsed from disconnection to complete drying. Two hydroperiod classes were defined for grouping the physicochemical and biotic data: quick-drying pools dried 1.5–32 h after disconnection and slow-drying pools dried 69–194 h after disconnection.

To ensure sample independence, the three rock baskets collected from each pool on each sampling date were treated as subsamples and the data averaged before analysis. Densities (individuals per rock basket) of ben-

thic invertebrates were grouped at the order level, and the relative abundance of different orders was compared in the connected and disconnected periods, and in the slow-drying and quick-drying hydroperiod classes. Emergence rates of aquatic insects were calculated as individuals trapped per day. The emergence rate data were subdivided into eight groups for analysis: Chironomidae, Ephemeroptera, cased (portable-case constructing) and uncased (free-living/web-spinning) Trichoptera, all Trichoptera, non-aquatic taxa, all insect taxa, the numerically dominant genus *Deleatidium* (Ephemeroptera, Leptophlebiidae) and the proportion of Ephemeroptera.

Two-sample *t*-tests were used to compare six invertebrate community variables calculated from the rock-basket data in connected and disconnected pools (both quick and slow drying): densities of the three most abundant orders (Diptera, Ephemeroptera and Trichoptera), total invertebrate density, invertebrate taxon richness and Shannon–Wiener diversity. Total invertebrate densities were square-root-transformed, and the remaining variables were log (*X*+1)-transformed to improve normality and variance homogeneity. Relationships between the six community variables and water levels in pools in the quick- and slow-drying hydroperiod classes were assessed by analysis of covariance (ANCOVA), using hydroperiod class as the categorical variable and water level as the covariate. Tests of homogeneity of slopes indicated that the hydroperiod class by water-level interaction was statistically significant for Shannon–Wiener diversity, and a linear regression of water level on Shannon–Wiener diversity was calculated for each of the two hydroperiod classes. A non-metric multidimensional scaling (NMDS) ordination was used to assess variation in composition of the benthos in response to three pool states (connected, disconnected and dry). The *t*-tests and ANCOVAs were run with SigmaPlot 12.5 (Systat Software, San Jose, CA, USA), and the NMDS was run with the function metaMDS from the vegan package in R (R Core Development Team).

Linear mixed-effects (LME) models were used to test relationships between insect emergence and change in water level in the scour pools. This allowed us to test for general relationships between drying and emergence, rather than focussing on separate relationships in individual pools. Model structure was determined with Akaike's information criterion (AIC), following the selection procedure outlined by Zuur *et al.* (2009). Separate models were run for four response variables: log-transformed total insect emergence, log (*X*+1)-transformed Ephemeroptera emergence, log (*X*+1)-transformed *Deleatidium*

spp. emergence and arcsine-transformed proportions of Ephemeroptera in total insect emergence. These response variables were selected on the basis of the composition of the emergence-trap samples; there were too few samples containing Diptera, Trichoptera and non-aquatic insect taxa to test relationships between emergence in these groups and pool water level. The LME models were fitted with random pool effects (intercepts, slopes) and fixed predictor effects so that pool-specific responses varied randomly around the general X–Y relationship, assuming a normal distribution. The models for total insect emergence rates and emergence rates of Ephemeroptera and *Deleatidium* spp. used random intercepts, and the model for proportions of Ephemeroptera in total insect emergence used random slopes and intercepts. The analyses were carried out using the lmer package in R version 2.9.1 (R Core Development Team).

Results

Hydrology and physicochemical conditions

The ephemeral section of the Selwyn River, including the study reach, flowed continuously for 7 months before the study began. This extended flow period began with a flood in February 2008, and scour pools in the study reach were connected from February 2008 until 10 November 2008, when four of the 10 pools disconnected. The remaining pools disconnected over the next 19 days and remained disconnected for the rest of the study. Pool hydroperiods following disconnection ranged from 1.5 to 194 h (Table 1); five pools were in the quick-drying hydroperiod class and five were

slow-drying. Despite the large-scale longitudinal gradient in intermittence in the upper Selwyn River, there was no regular longitudinal pattern in pool hydroperiods in the study reach.

Abrupt changes in the magnitude and temporal pattern of several physicochemical variables occurred when scour pools were hydrologically disconnected (Fig. 3). There were gaps in the time series for some pools due to logger failures, but the results shown in Fig. 3 for two complete data sets were typical of other complete and partial data sets. In both slow- and quick-drying pools, the amplitude of diel fluctuations in water temperature increased after disconnection, and this continued until the pools dried (Fig. 3a,b). Temporal patterns in DO, pH and EC were similar across the slow-drying pools; DO concentration, pH and the amplitudes of diel DO and pH fluctuations all decreased after disconnection, and EC gradually increased (Fig. 3c,e). No gradual, monotonic changes in DO, pH or EC were apparent in quick-drying pools, which may be due to the short hydroperiods (Fig. 3d,f). Despite the large differences in temporal patterns in slow- and quick-drying pools, time-averaged values for most physicochemical variables were similar. The exception to this was maximum daily temperature, which was on average 37% higher in quick-drying pools during the disconnected period (Table 1).

The benthos

Aquatic and semi-aquatic invertebrates from 49 taxa occurred in the rock-basket samples. The most abundant taxa were Chironomidae (mean density: 56 per sample), Copepoda (30), *Deleatidium* spp. (18), *Olinga feredayi* (17)

Table 1 Mean values and ranges (in brackets) of physical and chemical conditions in connected and disconnected pool states in slow-drying and quick-drying scour pools

	Slow-drying		Quick-drying	
	Connected	Disconnected	Connected	Disconnected
Max pool depth (mm)	N/A	568 (350–750)	N/A	530 (300–950)
Hydroperiod (hours)	N/A	131.1 (81–193.75)	N/A	14.2 (1.5–32.25)
Dissolved oxygen daily average (% saturation)	113.2 (85.4–128)	97.9 (34.3–126.9)	116.8 (106.8–127.3)	106.1 (56.1–127.9)
Dissolved oxygen daily maximum (% saturation)	125.7 (93.7–150.8)	117.9 (43.2–150.8)	127.0 (114.1–151.0)	137.7 (118.9–157.1)
Dissolved oxygen daily minimum (% saturation)	101.4 (79.2–122.3)	74.4 (21.5–115.1)	106.8 (92.9–118.2)	80.2 (4.7–115.6)
Temperature daily average (°C)	14.4 (10.2–20.5)	15.4 (12.2–17.3)	16.5 (12.4–20.9)	18.5 (16.8–20.2)
Temperature daily maximum (°C)	16.9 (11.8–22.0)	18.9 (12.6–32.2)	19.3 (15.1–25.7)	29.9 (23.3–36.6)
Temperature daily minimum (°C)	12.4 (8.1–20.2)	12.9 (4.9–16.2)	14.3 (9.3–20.4)	12.1 (8.5–15.7)
Electrical conductivity daily average ($\mu\text{S cm}^{-1}$)	85.2 (70.1–200.1)	85.2 (31.4–101.1)	85.4 (74.3–121.1)	58.4 (12.0–102.7)
Electrical conductivity daily maximum ($\mu\text{S cm}^{-1}$)	90.6 (77.5–202.2)	92.3 (76.1–111.9)	97.7 (85.1–200.0)	101.6 (82.2–112)
Electrical conductivity daily minimum ($\mu\text{S cm}^{-1}$)	80.9 (71.0–198.2)	79.0 (13.4–96.6)	79.5 (73.7–86.9)	88.1 (insufficient data)
pH average	7.3 (5.7–11.4)	7.3 (5.7–11.4)	7.6 (7.4–8.04)	7.4 (6.9–7.8)
pH maximum	7.7 (6.6–10.1)	7.3 (5.7–11.3)	7.8 (7.6–8.5)	8.3 (7.4–9.1)
pH minimum	7.8 (6.9–8.2)	6.9 (4.5–7.5)	7.5 (7.2–7.7)	6.5 (4.2–7.7)

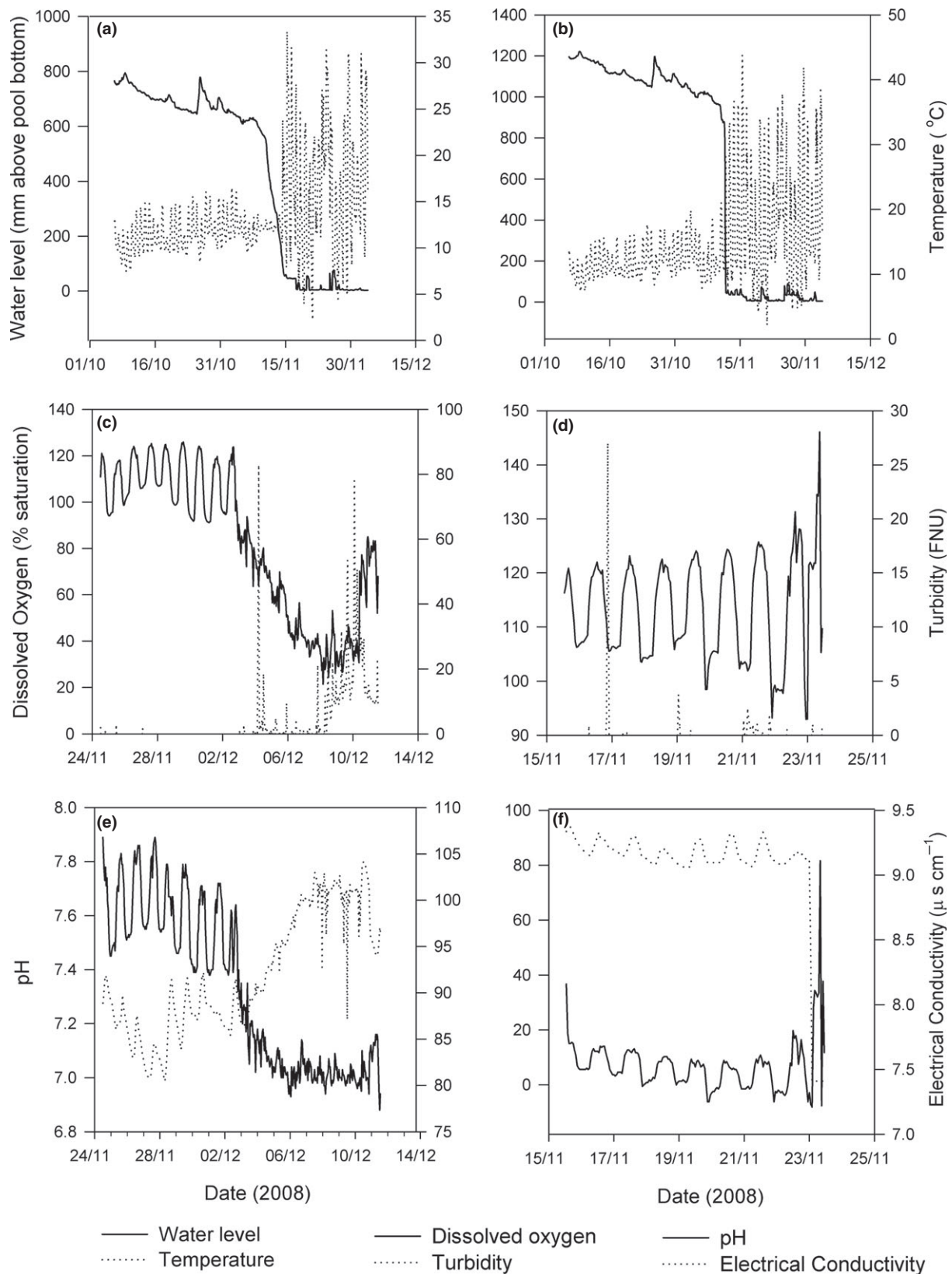


Fig. 3 Changes in physicochemical conditions from October to December 2008 in slow-drying (a, c, e) and quick-drying (b, d, f) pools. Date ranges in panels c–f are reduced to make changes near the drying date more apparent. Arrows indicate the time of pool disconnection. Last measurement dates shown correspond to dates of complete pool drying. Note changes in variable scales and date ranges.

and *Neozephlebia scita* (11). The most abundant insect orders were Ephemeroptera, Diptera and Trichoptera. Pool disconnection was associated with statistically significant changes in several community variables, as indicated by paired comparisons between the connected and disconnected periods (Fig. 4). The mean density of Ephemeroptera larvae increased, and the mean density of Diptera larvae decreased between connected and disconnected states ($t = -4.54$, $P = <0.001$ and $t = 2.29$, $P = 0.02$, respectively; Fig. 4a). Diptera were dominated by Chironomidae in both connected (98.9% Chironomidae) and disconnected (99.4% Chironomidae) states. Mean taxon richness and mean Shannon–Wiener diversity increased in disconnected states ($t = -6.15$, $P < 0.001$ and $t = -5.15$, $P < 0.001$, respectively; Fig. 4c, d). Differences between the mean densities of Trichoptera larvae and total invertebrates in connected and disconnected states were not statistically significant ($P > 0.05$ for each comparison) (Fig. 4a,b).

The invertebrates in rock-basket samples were dominated by univoltine taxa (65% of taxa), with an average minimum larval period of 8 months (see Supporting Information Table S1). All Diptera taxa in the samples were plurivoltine, with relatively short larval and pupal periods (weeks to months and days to weeks, respectively). Most of the Ephemeroptera taxa in the samples were univoltine or bivoltine, with larval periods of 6–11 months. *Deleatidium* was an exception; this genus has a short summer cohort in New Zealand, with a larval development time of 3–4 months (Huyrn, 1996; Greenwood & McIntosh, 2004). With the exception of the plurivoltine *Oxyethira albiceps* and *Paroxyethira* sp., Trichoptera in the rock-basket samples were univoltine, with larval and pupation stages of 6–10 months and 2–4 weeks, respectively. Approximately 10% of the taxa in the rock-basket samples used atmospheric oxygen, 84% of the taxa had an aerial adult, 4% had behavioural traits that may enhance resistance (e.g. burrowing), and

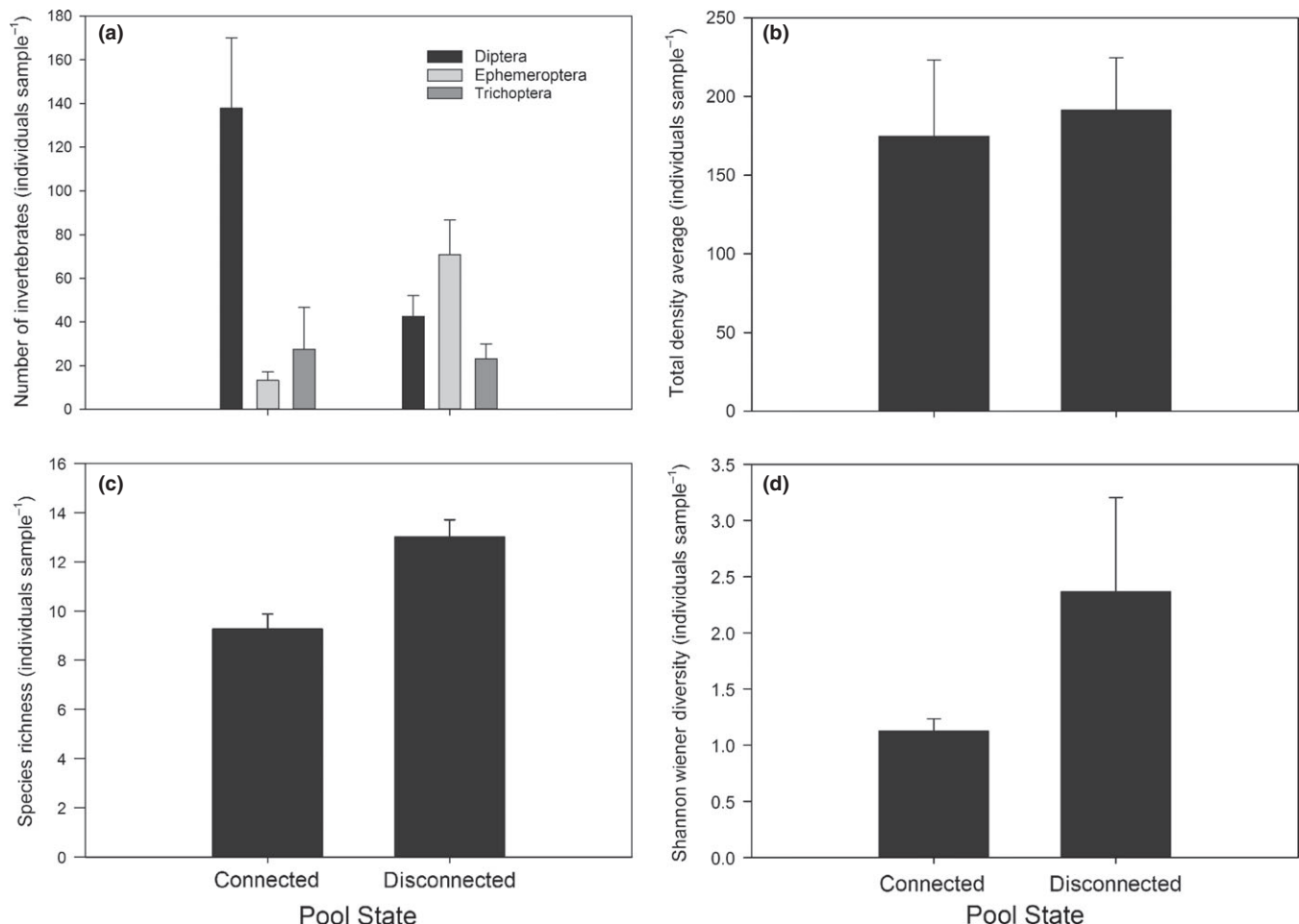


Fig. 4 Invertebrate assemblages in scour pools during connected and disconnected periods; values shown are for the combined data from 10 scour pools. Values are mean sums + 1 standard error).

22% had chitinous exoskeletons or external cases which increase desiccation resistance (e.g. cased Trichoptera, Crustacea, Gastropoda) (see Table S1). Only four taxa are known to have flexible life histories in terms of maturation time, *Olinga feredayi* (Trichoptera), *Deleatidium* spp. (Ephemeroptera), *Austrosimulium* sp. (Diptera) and the family Chironomidae (Diptera) (see Table S1).

ANCOVAs were used to determine the effects of hydroperiod and water level on five of the six invertebrate community variables (densities of Diptera, Ephemeroptera and Trichoptera, total density and taxon richness) (Table 2). Densities of Ephemeroptera and Diptera were affected by changing water levels, but not by hydroperiod (Table 2; Fig. 5a,b). No effects of water-level change or hydroperiod on Trichoptera density and total invertebrate density were detected (Table 2, Fig. 5c,d). Taxon richness was significantly higher in quick- than in slow-drying pools (Table 2; Fig. 5e). ANCOVA was not used for Shannon–Wiener diversity as the homogeneity of slopes test indicated that the hydroperiod by water-level interaction term was statistically significant ($F_{1,39} = 4.35$, $P = 0.04$). Individual linear regressions indicated that Shannon–Wiener diversity increased as water level decreased in slow-drying pools ($F_{1,27} = 7.28$, $P = 0.012$, $R^2 = 0.22$; Fig. 5f), but there was no detectable effect of

water level on Shannon–Wiener diversity in quick-drying pools ($F_{1,14} = 0.41$, $P = 0.53$; Fig. 5f).

The NMDS ordination of rock-basket samples indicated that large changes occurred in the invertebrate assemblage as pools dried (Fig. 6a). Samples from connected, disconnected and dry pools were separated along Axis 1. Larval Diptera, larvae of free-living caddisflies (e.g. *Aoteapsyche colonica* and *Hydrochorema* sp.) and some larval mayflies (e.g. *Coloburiscus humeralis*) were common in samples from connected pools (Fig. 6b). Ephemeroptera (especially *Deleatidium* spp.) and cased Trichoptera were dominant in samples from disconnected pools, and predatory insects (e.g. *Stenoperla prasinata*, *Archichauliodes diversus*) were also present (Fig. 6b). Samples from dry pools were dominated by terrestrial and desiccation-resistant aquatic invertebrates (e.g. spiders, mites, collembolans, copepods, adult Diptera; Fig. 6b). Larval Diptera were uncommon in disconnected pools that were almost dry. This observation, and the decline in density of larval Diptera as water level declined (Figs 4a & 5b), suggests that rapid emergence following disconnection depleted Diptera populations.

Emergence

The means and ranges of emergence rates (in individuals $\text{trap}^{-1} \text{day}^{-1}$) across the eight scour pools were 24.0 (range 2.0–98.3), 2.8 (range 0–15.7) and 2.3 (range 0–14.3) for emergence of total insects, Ephemeroptera and *Deleatidium* spp., respectively. The mean proportion of Ephemeroptera in samples was 0.17% (range 0–0.70).

Emergence rates of Trichoptera and non-aquatic taxa, including Elmidae, Staphylinidae and Mesoveliidae, increased substantially after scour pools became disconnected (Table 3); some of the non-aquatic taxa may have crawled into the emergence traps from adjacent dry land. Mean emergence rates of Trichoptera and non-aquatic taxa increased by 75 and 433%, respectively. Cased Trichoptera were more abundant in the emergence-trap samples than non-cased Trichoptera, and the mean emergence rate of cased Trichoptera increased by 205% after disconnection. Mean emergence rates of Ephemeroptera and Chironomidae decreased after disconnection by 28 and 231%, respectively (Table 3). However, the mean proportion of Ephemeroptera in total insect emergence increased by 45% after disconnection.

Linear mixed-effects (LME) models of emergence in response to change in pool water level were estimated for total insect emergence and variables related to Ephemeroptera, but not for the other community variables due

Table 2 ANCOVA results for tests of the effects of hydroperiod class and water level on invertebrate community variables, with homogeneity of slopes tests shown as hydroperiod class $\times$ water level interaction. Hydroperiod classes were categorical variables and pool water levels were covariates

Dependant variable	Test	Degrees of freedom	F	P
Ephemeroptera	Hydroperiod class	1,40	0.01	0.926
	Water level	1,40	4.65	0.037
	Hydroperiod class $\times$ water level	1,39	0.52	0.476
Diptera	Hydroperiod class	1,40	1.46	0.234
	Water level	1,40	15.42	<0.001
	Hydroperiod class $\times$ water level	1,39	0.70	0.407
Trichoptera	Hydroperiod class	1,40	1.63	0.209
	Water level	1,40	0.29	0.593
	Hydroperiod class $\times$ water level	1,39	0.65	0.425
Total invertebrate density	Hydroperiod class	1,40	0.70	0.407
	Water level	1,40	0.37	0.546
	Hydroperiod class $\times$ water level	1,39	0.14	0.707
Taxon richness	Hydroperiod class	1,40	4.39	0.042
	Water level	1,40	0.404	0.528
	Hydroperiod class $\times$ water level	1,39	0.001	0.982

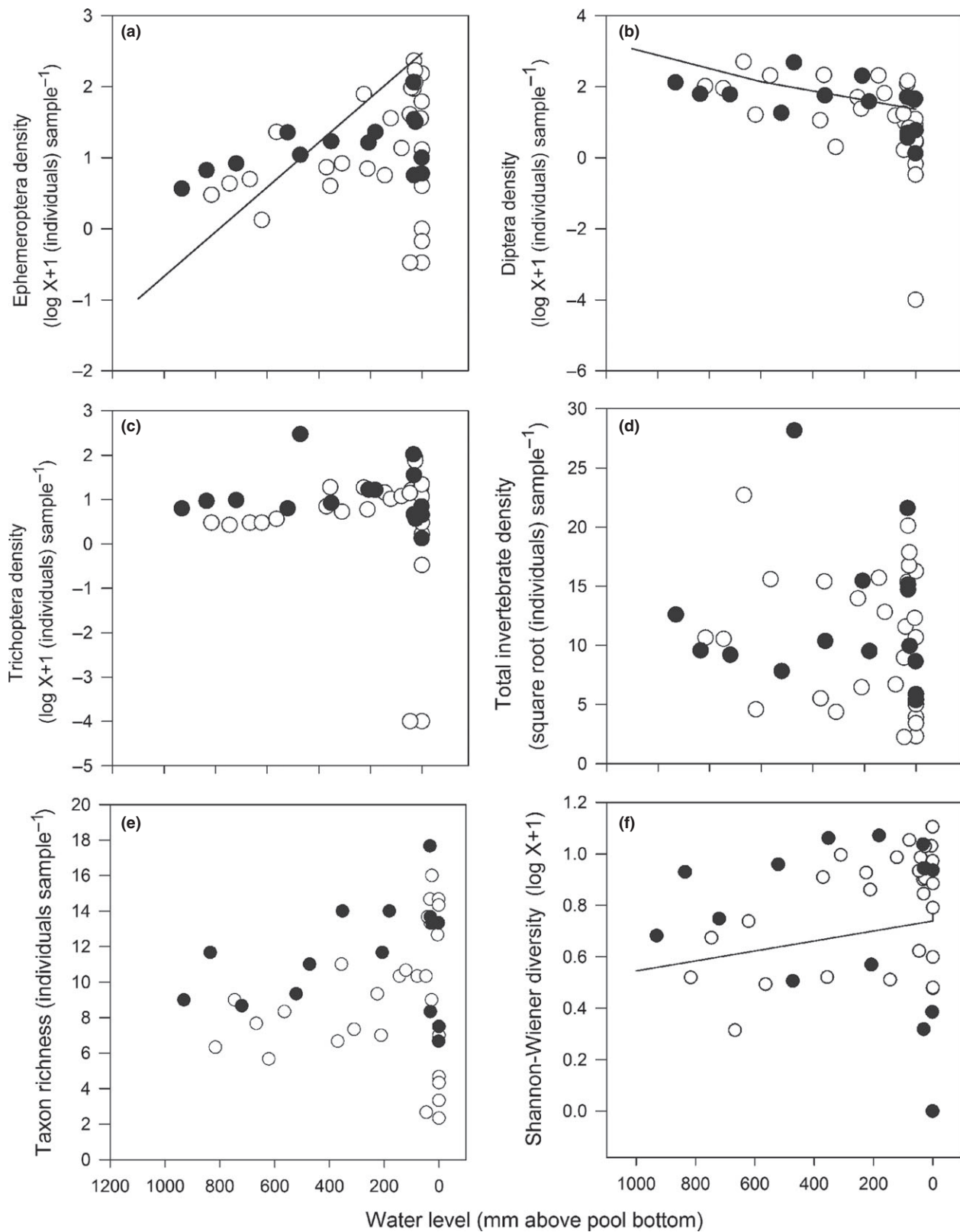


Fig. 5 Responses of macroinvertebrate community variables to declining water level in slow-drying (open circles) and quick-drying (black circles) pools. Lines in (a) and (b) indicate ANCOVA slopes and intercepts for significant relationships with water level, across hydroperiod class. Line in panel f indicates linear regression for slow-drying pools only. Each point represents the mean from three rock-basket samples collected on one date, and x-axis is shown in reverse order to represent pool drying.

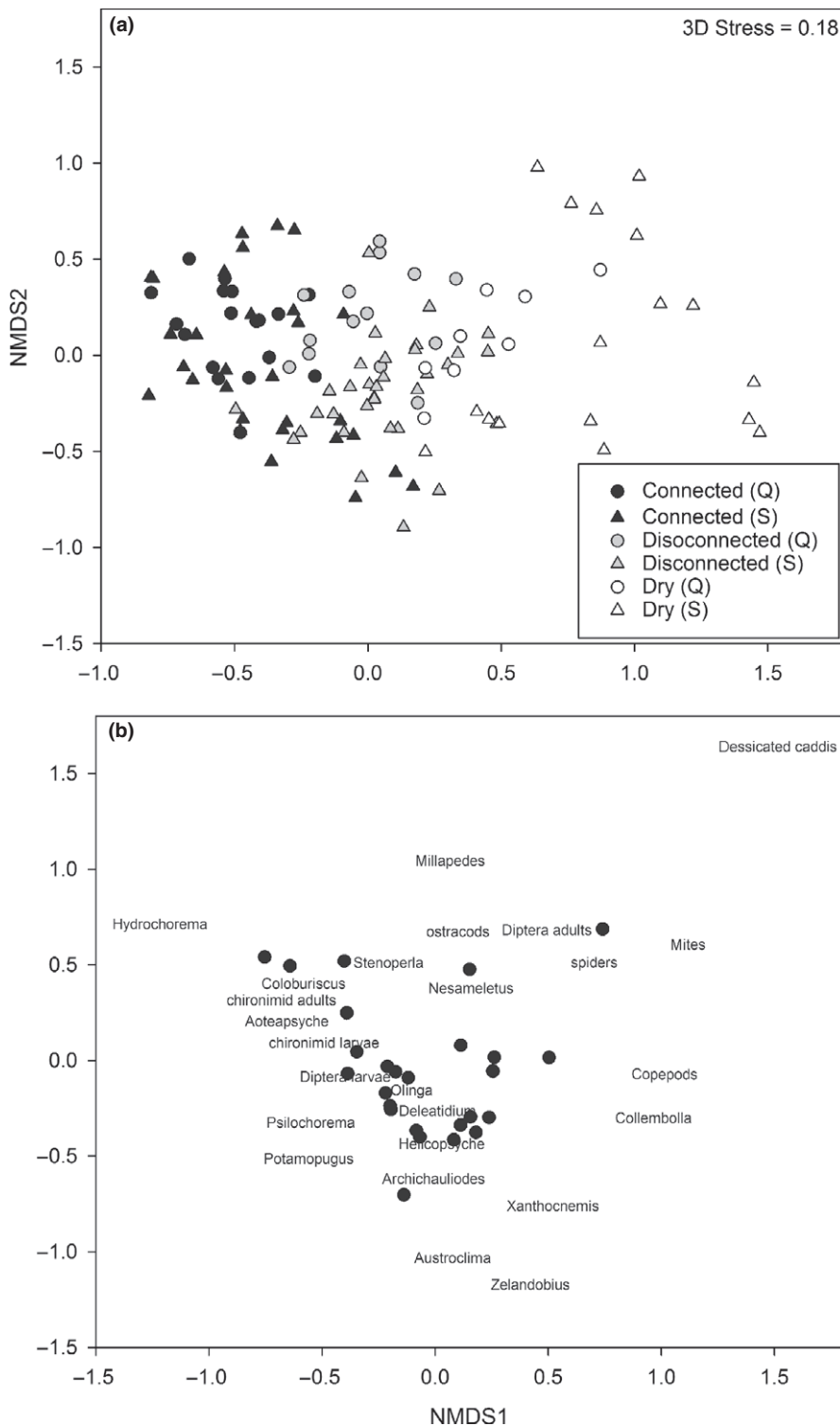


Fig. 6 (a) Plot of non-metric multidimensional scaling (NMDS) of benthic macroinvertebrate samples from scour pools. (b) Taxon scores from the NMDS.

to insufficient data to run the model. Total insect emergence rates varied monotonically as water level declined in each scour pool; positive and negative relationships were observed for different pools in both hydroperiod classes (Fig. 7). The LME models were also used to identify patterns of emergence across the scour pools by

partitioning inter-pool variation, as it was high (Fig. 7). The models indicated that total insect emergence across pools gradually increased as water level declined (Table 4; Fig. 8a) and that total Ephemeroptera and *Deleatidium* spp. emergence decreased as water level declined (Table 4; Fig. 8b,c). The proportion of Ephemeroptera in

Table 3 Mean emergence rates (number day⁻¹) and standard deviation (in parentheses) of numerically dominant taxa (>5% of individuals sampled), non-aquatic insects and all insects combined in connected and disconnected pools

Taxa	Connected	Disconnected
Chironomidae	17.16 (4.46)	5.18 (1.63)
Ephemeroptera	3.02 (0.80)	2.36 (0.33)
<i>Deleatidium</i> spp.	2.39 (0.67)	2.25 (0.29)
All Trichoptera	0.70 (0.21)	1.23 (0.45)
Uncased Trichoptera	0.43 (0.18)	0.39 (0.18)
Cased Trichoptera	0.27 (0.11)	0.83 (0.36)
Non-aquatic taxa	0.94 (0.31)	5.01 (2.67)
All insects	25.22 (4.99)	21.76 (4.04)

samples did not vary systematically in response to change in water level (Table 4; Fig. 8d).

Discussion

Previous studies of the responses of aquatic invertebrates to flow recession and drying have focussed on benthic community succession in habitats that dry slowly (weeks to months) (Boulton, 2003; Acuña *et al.*, 2005; Bonada *et al.*, 2007). The effects of rapid drying (hours to days) on the benthic community and on insect emergence have not previously been examined. In rivers with high rates of flow loss due to seepage and evaporation, and in highly ephemeral rivers that rarely flow, a substantial proportion of the channel bed is characterised by short hydroperiods (Larned, Datry & Robinson, 2007; Fritz, Johnson & Walters, 2008; Jacobson & Jacobson, 2012). In these rivers, invertebrate responses to rapid habitat change are likely to affect community structure in both the short term (e.g. around the scale of a generation) and in the longer term (e.g. at successional scales).

Results of this study and of previous research in the same river reach (Corti *et al.*, 2011) suggest that ecological conditions can vary widely across the short-hydroperiod (rapidly drying) habitats in the river channel. Some of this ecological variation is attributable to variable rates of habitat contraction, which in turn affects water chemistry and temperature, organic matter dynamics, the abundance and composition of the invertebrate assemblage, and the proportions and composition of inhabitants that emerge. The effects of declining water level on the invertebrate benthos are modified by several additional factors, including developmental modes of the taxa in pools, and the presence or absence of taxa with traits conferring resistance to desiccation. These factors constrain potential responses of invertebrates to drying in terms of escape and resistance.

The prediction that holometabolous taxa would dominate the assemblage remaining in near-dry pools was not supported. Instead, a large proportion of the holometabolous Diptera appeared to complete development and emerge prior to drying, as indicated by rapid declines in both benthic density and emergence rates after disconnection. Other factors, such as predation, could account for part of the decline in benthic Diptera, although fish were absent in most scour pools after disconnection (personal observation). The most common Diptera in the scour pools were Chironomidae and *Austrosimulium* sp.; both groups are multivoltine and some species in these groups have rapid development and a short pupal stage (Hunter, 1978; McKie, Cranston & Pearson, 2004). Chironomidae are the dominant macroinvertebrates in many intermittent aquatic habitats and comprise most of the 'specialist taxa' that occur exclusively in intermittent rivers (de Moor, 1989; Frouz *et al.*, 2003; Bogan, Boersma & Lytle, 2013). The time constraints imposed by holometabolous development in Chironomidae and *Austrosimulium* sp. did not prevent emergence in our study reach because the time required to complete development can be shorter than the hydroperiods of some scour pools (maximum 8 days). Higher taxonomic resolution of Chironomidae may have provided additional information on species-specific developmental restrictions in these habitats.

In comparison with Diptera, the development times for most Trichoptera (holometabolous) and Ephemeroptera (hemimetabolous) in the Selwyn River scour pools appeared to exceed the pool hydroperiods. New Zealand Trichoptera are predominately univoltine and have larval periods of c. 10 months, far longer than the hydroperiods of the scour pools. In contrast, many Ephemeroptera have rapid and/or flexible development and larval stages as short as 8–14 days (Gray, 1981). *Deleatidium*, the dominant Ephemeroptera genus in many New Zealand streams, has a flexible, bivoltine life history with a fast-growing summer generation and a slow-growing winter generation (Winterbourn, 1974; Scrimgeour, 1991; Greenwood & McIntosh, 2004; Robinson & Buser, 2007). Due to the asynchronous development of two *Deleatidium* cohorts, a range of size classes would have been present in the Selwyn River scour pools after disconnection. Only individuals that had reached a minimum size threshold before pool disconnection would have been able to mature and emerge during the short hydroperiods. Clearly, our prediction that hemimetabolous taxa would dominate emergence samples following pool disconnection needs to be refined, which will require more information about developmental patterns in aquatic insects. In contrast to the

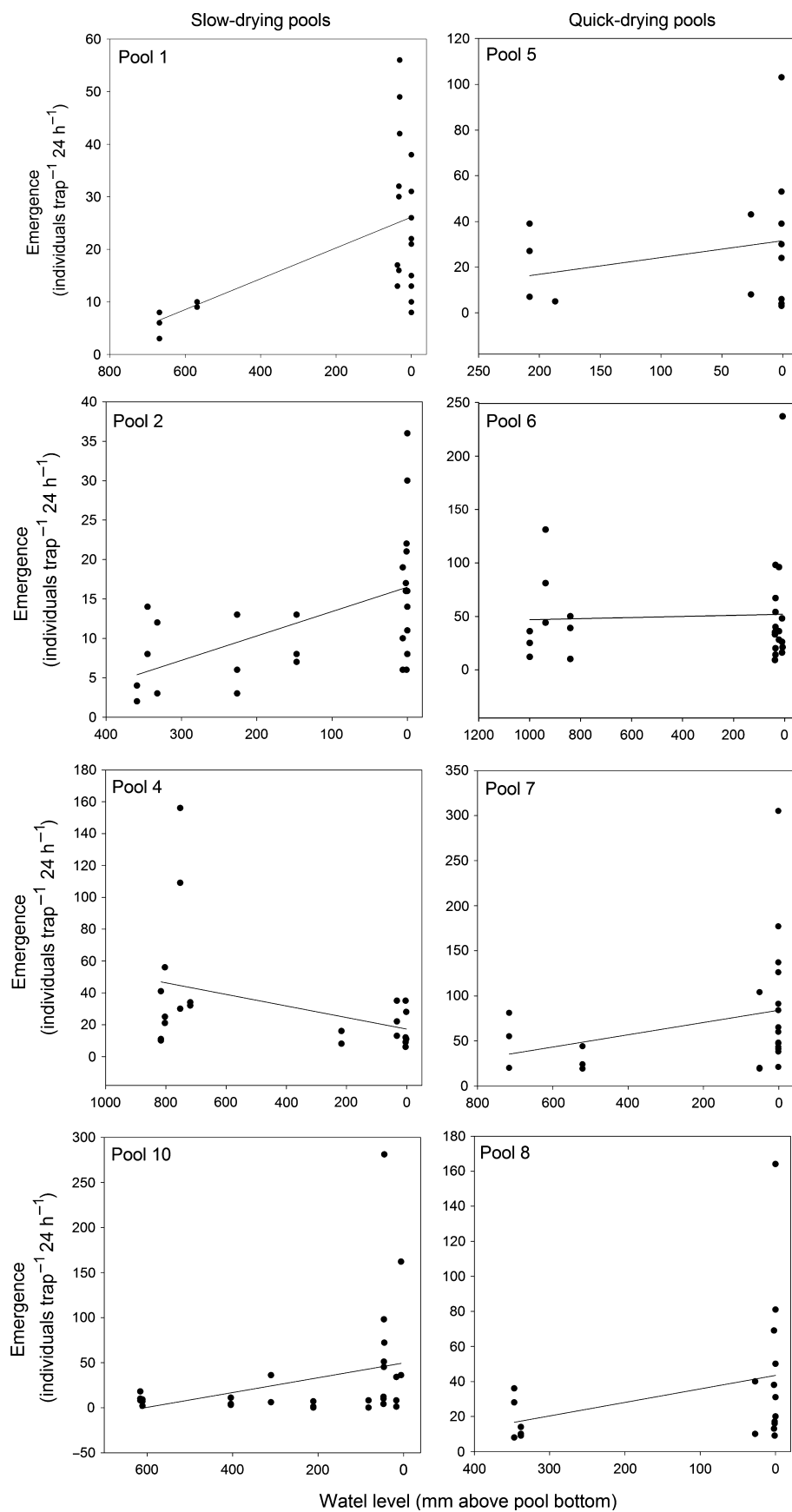
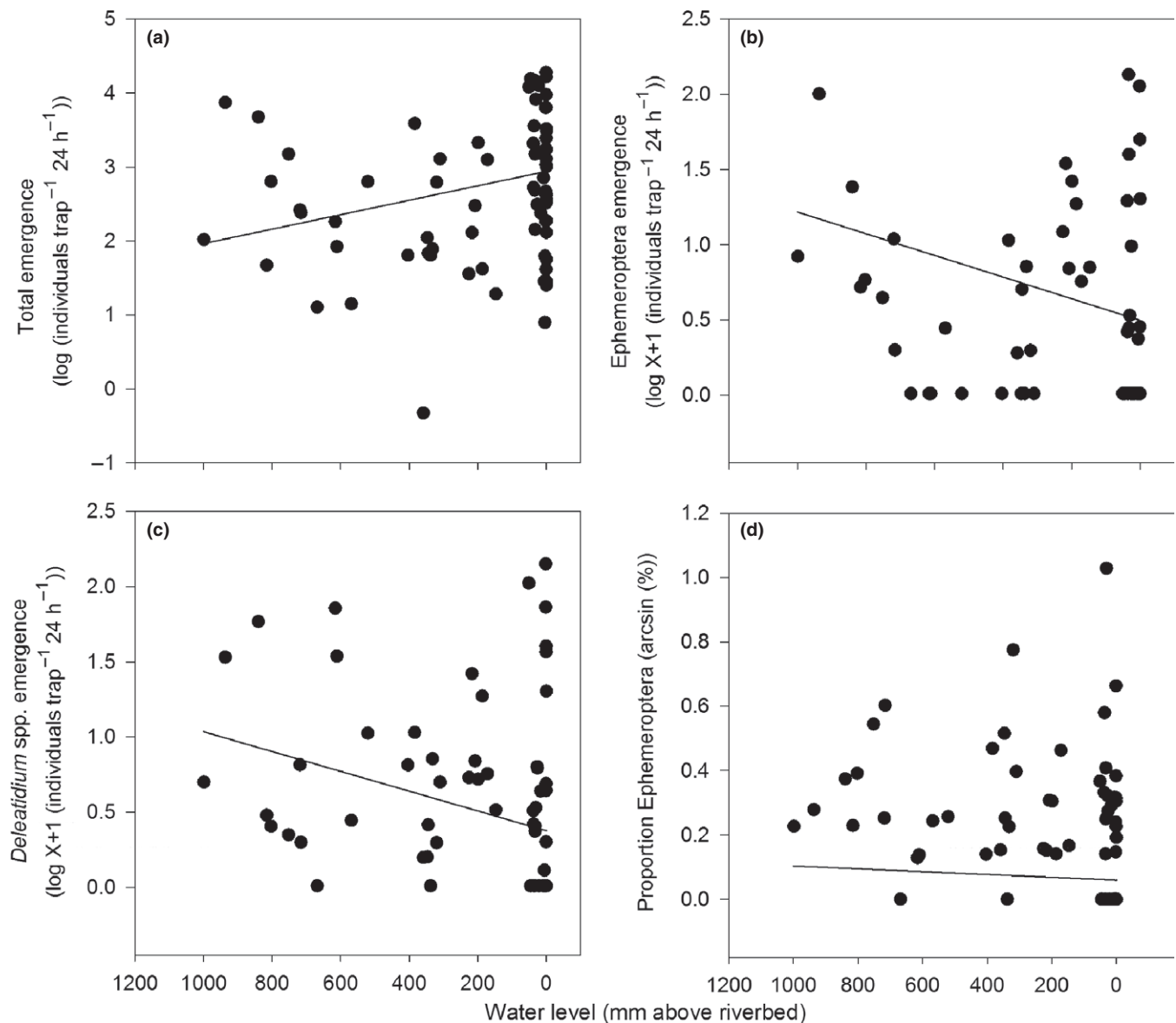


Fig. 7 Variation in total macroinvertebrate emergence rates in individual scour pools as water level declined. Lines correspond to linear regression equations. Pool numbers correspond to the locations on Fig. 2.

Table 4 General linear mixed-effects model results for insect emergence variables. Model structures shown were chosen with Akaike's information criterion (AIC)

Response variable	Predictor	Transformation	Random effects structure	Significance of predictor			Parameter estimate	
				Likelihood ratio test	<i>P</i>	<i>t</i>	Intercept	Slope
Total insect emergence	Water level	Natural log(<i>x</i>)	Random intercepts	8.1	0.004	-2.58	2.94	-0.001
Ephemeroptera emergence	Water level	Natural log(<i>x</i> +1)	Random intercepts	7.1	0.007	2.74	0.49	0.001
<i>Deleatidium</i> spp. emergence	Water level	Natural log(<i>x</i> +1)	Random intercepts	7.6	0.005	2.81	0.37	0.001
Proportion Ephemeroptera emergence	Water level	Arcsine	Random slopes and intercepts	3.5	3.559	2.01	0.06	0.003

**Fig. 8** Insect emergence rates in response to declining water level across scour pools, with significant trend lines from linear hierarchical models.

well-studied genus *Deleatidium*, detailed life-history information is currently lacking for most aquatic insects in New Zealand's intermittent rivers. This information is

gradually emerging, and refined predictions about population- and community-level responses to flow intermittence will be possible in the future (Storey & Quinn, 2011).

Aquatic invertebrates that lack an aerial adult or have prolonged development may survive dry periods by resisting desiccation *in situ* (Storey & Quinn, 2011). Approximately 60% of the taxa we collected had morphological and/or behavioural traits associated with desiccation resistance. These traits include respiratory spiracles and burrowing behaviour, opercula, exoskeletons and external cases that retain moisture (Winterbourn, 1970; Zamora-Muñoz & Svensson, 1996). The operculate gastropod *Potamopyrgus antipodarum*, which was present in our samples, survives for weeks to months out of water (Storey & Quinn, 2011). Other taxa in our samples that are considered desiccation resistant include Collembola, Oligochaeta, Nematoda and some Chironomidae. These taxa can encyst, aestivate, pupate or migrate into the bed to resist drying, or survive as active larvae. Live specimens of these taxa have been collected in the sediments of the Selwyn River and other rivers >200 days after drying (Dahms, 1995; Larned *et al.*, 2007).

Changes in the densities of different taxa in the scour pools as they dried affected benthic taxon richness but not Shannon–Wiener diversity. In previous studies of invertebrates in pools of intermittent rivers, taxon richness decreased after disconnection, due to the loss of sensitive taxa and the cessation of colonisation (Boulton & Lake, 1990; Bonada *et al.*, 2006). In the present study, taxon richness did not change continuously in response to changing water level, but was higher in quick- than in slow-drying pools. Higher taxon richness in quick-drying pools was not caused by colonisation of obligate lentic taxa, as the invertebrate assemblages in the scour pools were subsets of the lotic taxon pool that occurred throughout the river mainstem (Arscott *et al.*, 2010). Lentic taxa are very uncommon in the Selwyn River and other highly intermittent alluvial rivers (Datry *et al.*, 2014). The scarcity of lentic invertebrates in isolated pools in these rivers is presumably because the time required for their dispersal, oviposition, hatching and growth exceeds the lifetime of the pool (Arscott *et al.*, 2010). The higher taxon richness that we observed in quick-drying pools was probably due to the concentration of taxa, as invertebrates quickly moved down the pool sides ahead of the declining water surface, and into the rock baskets. These observations suggest that different mechanisms control responses of the benthos to drying in aquatic habitats with long and short hydroperiods. Long-hydroperiod habitats may be characterised by colonisation and, as drying progresses, by the replacement of taxa sensitive to desiccation and to declining water quality by more tolerant taxa. In these

habitats, we expect (i) lentic taxa, tolerant of hypoxia, to replace lotic taxa following pool disconnection, (ii) desiccation-tolerant semi-aquatic taxa (e.g. collembola, staphylinid beetles) to replace lentic taxa during pool drying and (iii) terrestrial taxa to replace semi-aquatic taxa in dry pools (Boulton & Lake, 2008). In contrast, assemblages in short-hydroperiod habitats are likely to shift directly from lotic to terrestrial taxa, with little colonisation by lentic and semi-aquatic taxa. The temporal patterns we observed in samples from rock baskets may have been affected by sampling baskets from the end of a chain, rather than sampling randomly. However, the likelihood of substantial position effects is small and random sampling would have required more handling and caused inadvertent disturbances.

The study of the ecology of intermittent rivers is growing rapidly, partly in response to predicted increases in flow intermittence due to climate change and water abstraction (Larned *et al.*, 2010a,b). Community-level responses of aquatic invertebrates to flow intermittence are among the most common research topics in this field. Most of the invertebrate studies have focussed on sites with long hydroperiods (months to years) (e.g. Bogan & Lytle, 2011) or have used annually or seasonally averaged estimates of intermittence (e.g. Arscott *et al.*, 2010). To our knowledge, the current study is the first to focus on rapidly drying habitats (hours to days) in an unmodified river. Hydroperiods that are short relative to invertebrate lifespans or larval or pupal phases impose severe constraints on community responses. Unlike long-lived habitats, where community change is driven primarily by colonisation, biotic interactions and emigration, community changes in short-hydroperiod habitats are driven by emigration (particularly through emergence) and mortality. Models of community dynamics in intermittent aquatic systems generally assume that community dynamics are jointly controlled by physical (e.g. drying, heating, hypoxia) and biotic (e.g., predation, competition) processes (e.g. Schneider & Frost, 1996; De Meester *et al.*, 2005). Based on the results of the current study, we predict that the invertebrate assemblages in short-hydroperiod habitats of temporary rivers are almost entirely physically controlled. Three corollaries of this prediction are as follows: 1) initial invertebrate assemblages are composed of organisms trapped as pools become disconnected; 2) change during drying is determined by mortality and escape (by emergence or crawling); and 3) the assemblages that remain after complete drying consist of those organisms with behavioural, physiological and morphological traits that confer resistance to desiccation.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Invertebrate taxa in rock-basket (RB) and emergence-trap (ET) samples, and life history, behavioural and morphological related to desiccation resistance or escape from drying habitat.

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